

But this was not the stated purpose of its military designer.

The laser is a prime illustration of a tool made for a given purpose (actually no real purpose) that then found applications that were not even dreamed of at the time. It was a typical “solution looking for a problem.” Among the early applications was the surgical stitching of detached retinas. Half a century later, *The Economist* asked Charles Townes, the alleged inventor of the laser, if he had had retinas on his mind. He had not. He was satisfying his desire to split light beams, and that was that. In fact, Townes’s colleagues teased him quite a bit about the irrelevance of his discovery. Yet just consider the effects of the laser in the world around you: compact disks, eyesight corrections, microsurgery, data storage and retrieval—all unforeseen applications of the technology.*

We build toys. Some of those toys change the world.

Keep Searching

In the summer of 2005 I was the guest of a biotech company in California that had found inordinate success. I was greeted with T-shirts and pins showing a bell-curve buster and the announcement of the formation of the Fat Tails Club (“fat tails” is a technical term for Black Swans). This was my first encounter with a firm that lived off Black Swans of the positive kind. I was told that a scientist managed the company and that he had the instinct, as a scientist, to just let scientists look wherever their instinct took them. Commercialization came later. My hosts, scientists at heart, understood that research involves a large element of serendipity, which can pay off big as long as one knows how serendipitous the business can be and structures it around that fact. Viagra, which changed the mental outlook and social mores of retired men, was meant to be a hypertension drug. Another hypertension drug led to a hair-growth medication. My friend Bruce Goldberg, who understands randomness, calls these unintended side applications “corners.” While many worry about unintended consequences, technology adventurers thrive on them.

The biotech company seemed to follow implicitly, though not explicitly, Louis Pasteur’s adage about creating luck by sheer exposure. “Luck favors the prepared,” Pasteur said, and, like all great discoverers, he knew something about accidental discoveries. The best way to get maximal exposure is to keep researching. Collect opportunities—on that, later.

To predict the spread of a technology implies predicting a large element of fads and social contagion, which lie outside the objective utility of the technology itself (assuming there is such an animal as objective utility). How many wonderfully useful ideas have ended up in the cemetery, such as the